Storage in Electricity Distribution: Grids vs Markets

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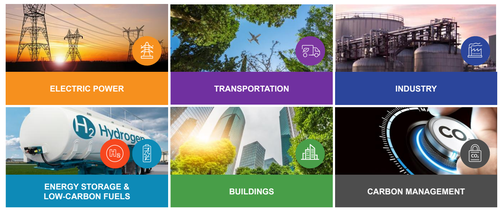
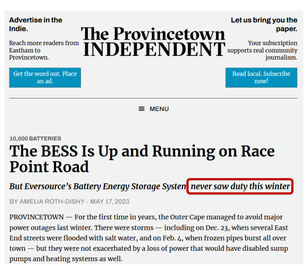


Future Energy Systems Center

Motivation

- Meeting growing electricity demand requires significant **distribution grid upgrades** costing billions of dollars in Massachusetts.
- Battery **storage can reduce peak demand** and the associated investments.
- In deregulated states, storage owned by distribution companies is typically not allowed to participate in electricity markets, which leads to **low utilization** (e.g., one cycle per month).
- States experiment with **allowing market participation to reduce retail prices**.
- In a New York pilot, yearly revenue from market participation accrues to **$8 per kWh** of battery capacity.



Objectives

- **Quantify the value of storage** for peak shaving (the **grid value**) and market participation (the **market value**).
- Formal description of **market participation constraints** for battery control optimization and investment planning.
- **Inform state policy** by comparing a numerical case study with the results from ongoing pilot projects.

Approach / Analysis

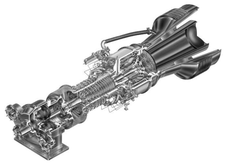
Supply Resources

We consider three sources of supply:

Grid

Storage

Backup Generation



Market Constraints

We consider two types of market participation:

- **Full participation**, meaning that all resources can supply electricity regardless of system conditions.
- **Peak shaving**, which only allows supply from storage and backup generation if grid supply is insufficient to meet demand.

Resource Optimization

We consider two optimization problems:

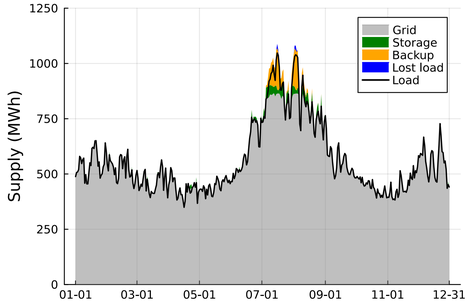
- **System operation**, which determines the operation of each resource under the market constraints for fixed resource capacities, electricity demand and prices.
- **Investment planning**, which determines future investments into each type of resource considering their future operation.

Both problems are solved with standard **mathematical programming** techniques.

Results (2024 Nantucket Case Study)

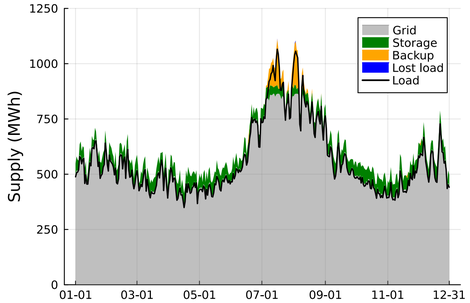
Peak Shaving Only (Status Quo)

- Storage is only used at times of peak demand and supplies 1.7 GWh.



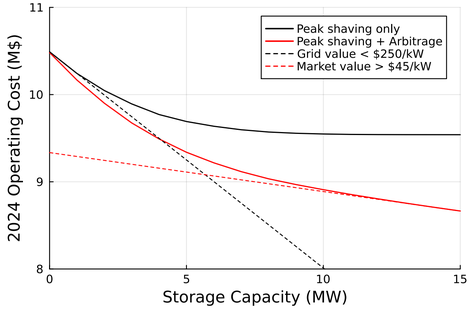
Peak Shaving + Market Participation (Arbitrage)

- Storage is used throughout the year and supplies 16.7 GWh.



Grid vs Market Value

- The marginal grid value decreases in storage size, the market value stays constant.



Takeaways

- We find a **market value** of $45/kW for an 8-hour battery, i.e., about **$6/kWh** of battery capacity.
- Realizing this value **increases battery utilization by a factor 10**.
- The **New York pilot reports similar values** and finds that increased wear and tear requires a **downtime of 1 day per month for maintenance**.
- Over a 20-year lifetime, the extra revenue amounts to $120/kWh, about **half the projected future cost of battery storage**.

Next Steps

Technical

- Incorporate the grid and market value into **investment planning**.

Practical

- Investigate how **distribution companies** think about storage and market participation.

Policy

- Investigate how **policy makers and regulators** think about distribution-grid storage and market participation.

References

- Electric Sector Modernization Plans by Eversource and National Grid.
- National Grid East Pulaski Energy Storage System Wholesale Market Participation Annual Report 2024.
- Pomona Battery Energy Storage System Annual Report 2024.
- Nantucket Island Energy Storage Assessment, PNNL, 2019.
- Cost Projections for Utility-Scale Battery Storage: 2023 Update, NREL, 2023.